

Team Meeting Minutes: September 22, 2026

2026-09-22 · @Someone

Meeting details

Field	Value
Date	September 22, 2026, about 3:10 PM
Length	About 17 minutes across two recordings (roughly 6 minutes, then 11)
Attendees	Student team: Tyler, Chris, Matthew, Joe (names as captured by transcription)
Sponsor	Dr. Qualls not present
Purpose	Weekly team check-in ahead of class

Deliverables due Thursday

The only item on the agenda was the value proposition and product requirements, both due Thursday and expected to be covered in class that day.

- Much of the content already exists in the Git repo documentation, which captures what the sponsor wants, the deliverables, and the overall expectation. It just is not formatted to the assignment outline.
- The value proposition is one paragraph and considered straightforward. The product requirements template looked more involved.
- The template asks for mechanical, thermal, and electrical inputs and outputs. This project is software only, so most of those sections do not apply. The team considered covering the UI as the closest fit.
- Requirements captured in the last sponsor meeting should carry over into the product requirements document, mainly reworded.
- The detailed notes from that sponsor meeting are posted in the discussion thread for that meeting.

Demo: pulling MODIS data with earthaccess

A team member screen-shared working Python code written for Mary's class, which pulls the same MODIS data the capstone will use.

- The script uses the `earthaccess` library to log in, then queries MODIS by short name and downloads granules with `earthaccess.download`.
- Test case: a 10-day span in January 2024 for a mountain near Coeur d'Alene, identified as Fernan Saddle.
- Files arrive as HDF, which is much harder to work with than CSV and needs special handling.
- Latitude and longitude have to be converted into MODIS projected coordinates in meters.
- Idaho is covered by four tiles, so a single tile is roughly a quarter of the state. A tile-bounds function reads a file to find its extent and check whether the target point falls inside, then narrows from there.
- The rest of the script sorts file paths, loops through them, reads and plots each one, with some checks along the way.
- Biggest obstacle so far: getting access to the data at all.

What the plotted images showed

- Yellow areas appeared to be mostly cloud, and white was cloud for certain. Both lined up closely with the visible cloud shapes.
- Snow values fall in the 0 to 100 range, with more purple meaning more snow cover.
- The team recalled Dr. Qualls preferring a threshold somewhere between 10 and 40.
- A pixel value of 250 came up and was flagged as unknown, with a guess that it means full cloud cover.

Two corrections

1. The value 250 does mean cloud. In the MODIS snow product (MOD10A1), the NDSI_Snow_Cover band uses 0 to 100 for NDSI snow values and reserves higher numbers as flags. The full key is worth coding around, not just 250:

Value	Meaning
0 to 100	NDSI snow value
200	Missing data
201	No decision
211	Night
237	Inland water
239	Ocean
250	Cloud
254	Detector saturated
255	Fill

2. Matrix columns are years, not days. The team debated whether each column is a day or a section of a tile. From Dr. Qualls's explanation in the previous meeting it is neither. Each year's daily images collapse into one first-day-of-land image, that image is flattened into a single column, and the matrix holds one row per pixel and one column per year. His 17-column example came from 17 years of data.

Project logistics

- A shared form or board is being set up so the team can track each document in one place, with the instructions alongside them, instead of documents piling up and duties overlapping.
- Tyler created the repo and posted the link in the general channel.
- Dr. Qualls's suggested starting point still stands: scaffold modular code that accepts a variable-sized matrix and runs PCA, or find the right Python library to do it, so the pipeline is ready before real data arrives. Nobody has started this yet.
- Visualization work already underway is separate from that generic matrix and PCA step.
- The meeting ended early so everyone could get to class.

Action items

Owner	Task	Due
Team	Write the value proposition (one paragraph) and the product requirements, reusing the repo documentation and the sponsor meeting requirements	Thursday
Team	Decide how to handle template sections that do not apply to a software-only project, and whether the UI covers the inputs and outputs question	Thursday
Unassigned	Scaffold modular code that takes a variable-sized matrix and runs PCA	Not set
Demo author	Research the remaining MODIS flag values and add handling for them	Not set
Team	Post the value proposition and product requirements templates in the chat	Not set
Team lead	Finish the shared document tracking form so duties do not overlap	Not set